

Point compute at the next scientific discovery.

Published evidence → testable hypotheses → experiments a scientist can inspect.

PROTOTYPE IN PROGRESS

OPEN CATEGORY / PROPOSED



Conceptual research map

Science is abundant. The next experiment is scarce.

40M+

biomedical citations and
abstracts
in PubMed

- 01 Evidence is spread across papers.
- 02 Plausible connections multiply.
- 03 Laboratory time is limited.

Source: NIH / NLM, About PubMed. Citations and abstracts; not our ingested corpus.

Built around a scientist's next decision.

WORKING USER HYPOTHESIS

**A computational biologist
in a small oncology team.**

“Which mechanism deserves
our next experiment?”

See the research direction

A tree of questions and alternatives

Audit the evidence

Source passages, code and results

Make a deliberate decision

Uncertainty and failures stay visible

Design implements keyboard focus, reduced motion and dark/light themes; persona interviews remain to be done.

A research loop with evidence at every step.

01

Read

Papers become
linked claims.

02

→ Connect

Agents explore
possible bridges.

03

→ Test

Python analyses
use public data.

04

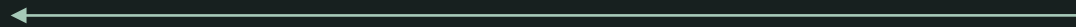
→ Challenge

A statistical screen
returns feedback.

05

→ Review

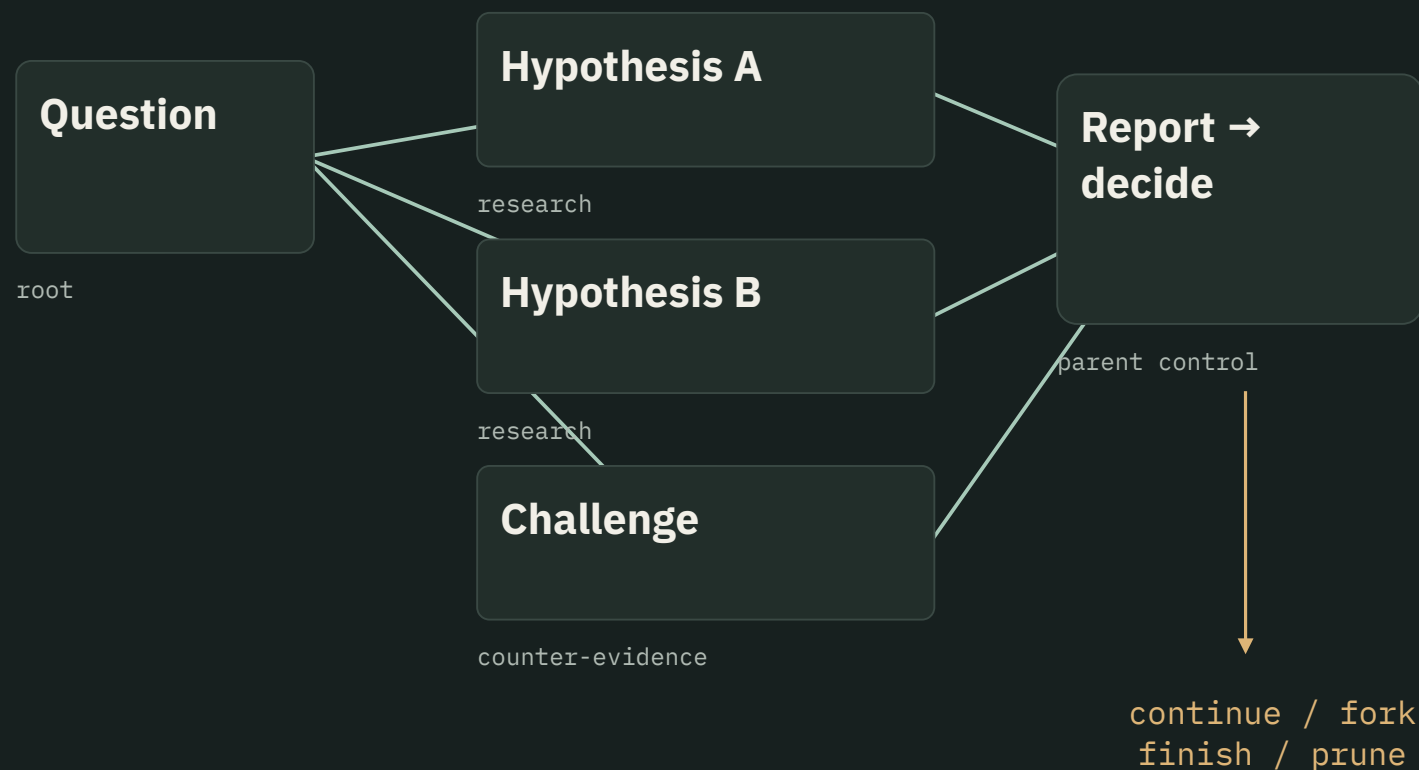
A scientist records
a decision.



Retry, reframe or pursue another branch

Knowledge graph = claims connected to entities, sources and context. Human review exists in the backend.

The engineering is in how the loop behaves.



Claims tied to sources

Exact quotes, identifiers and disagreements.

Explicit research actions

Method guidance, executable code and feedback.

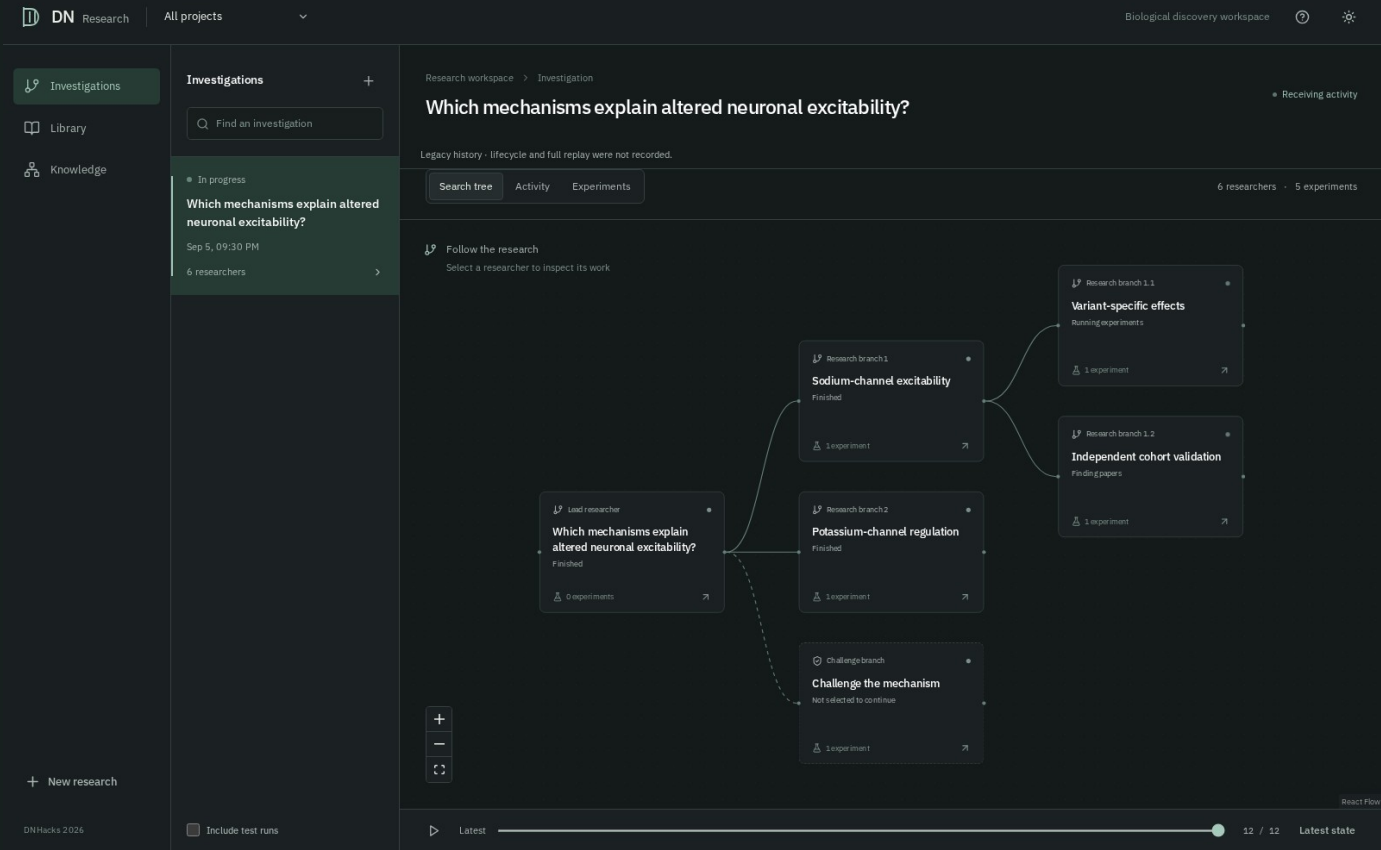
Research that resumes

Saved reports, decisions, events and artifacts.

Conceptual control diagram. Model judgment allocates research; calibrated statistical pruning is not active.

One question. An inspectable trail.

INTERFACE DEMO / SYNTHETIC DATA



PREPARED CORPUS

100
curated full-text
pancreatic cancer papers

Show source → hypothesis →
code → result → feedback

Replace this interface fixture with the final scientific run.

A plausible result must survive scrutiny.

01 / SOURCE

Did the paper say that?

Quoted spans, grounded IDs, biological context and explicit disagreements.

02 / SCREEN

Does the result pass the checks?

Typed result, reported sample size, direction, p-value and robustness.

03 / SCIENTIST

Is it worth pursuing?

Code and evidence remain reviewable. Human acceptance requires a written note.

Screened candidate ≠ independently reproduced result ≠ confirmed biological discovery.

We test the trust layer, too.

MEASURED / STANDALONE DIAGNOSTIC

10,000

synthetic no-difference runs

Separate training data. Frozen configuration.
Fresh pairs scored sequentially.

1.12%

false alarms in this simulation

112 / 10,000 ever crossed the threshold
95% interval 0.93–1.35% · nominal level 5%

Synthetic null audit of learned-encoder diagnostic. Not end-to-end discovery accuracy or a deployed stopping guarantee.

Start with one team and one research decision.

PROPOSED FIRST PILOT

Oncology research teams

User: computational biologist

Buyer: research lead / lab director

Workflow: target or mechanism
prioritization

Commercial hypothesis

Private research workspace + metered compute

Pilot, buyer and pricing model are hypotheses. No customer adoption, revenue or savings claimed.

01 **Reproducible**

Can another analyst rerun the result?

02 **Useful**

Would the expert act on the candidate?

03 **Economical**

What is the cost per reviewed candidate?

A focused prototype. A credible path forward.

NOW / IMPLEMENTED

Inspectable research engine

Grounded claim graph
Recursive research control
Code, results and replay

NEXT / PROPOSED

Evidence of useful discovery

Finish the real PDAC demo
Independent reruns
Matched-budget pilot

LATER / PLANNING

Deeper biological tools

Learned biological tests
Cellular ecosystems
3D experiment workbenches

PDAC = pancreatic ductal adenocarcinoma. Learned-tool plans and calibrated branch stopping remain separate future work.

More useful science from every dollar of compute.

Our ambition: strengthen the infrastructure
behind American scientific discovery.

NEXT MILESTONE

One independently reproduced, expert-reviewed candidate.



Every judging criterion has a place in the story.

MAIN PRIZES

25 **Problem & Real-World Impact**
SLIDES 02 · 06 · 11

50 **Technical Execution**
SLIDES 04-08

25 **Feasibility & Deployment**
SLIDES 09-10 · 15

BEST USE OF AI

25 **Novelty & Importance of AI**
SLIDES 04-05 · 16

50 **Technical Sophistication**
SLIDES 05 · 08 · 14

25 **Reliability, Evaluation & Trust**
SLIDES 07-08 · 13-14

BEST IN DESIGN

30 **Design Quality & Craft**
SLIDES 03 · 06

40 **User Experience & Usability**
SLIDES 03 · 06

30 **Understanding of the User**
SLIDES 03 · 09

Weights transcribed from the user-provided judging rubric. This is a coverage map, not a claimed score.

What is built, prepared and still to be proven.

BUILT	Claims, quotes and context	Grounding failures are deferred; source count is not proof.
BUILT	Research actions and branch control	Model judgment; no calibrated scientific stopping.
BUILT	Statistical screen + backend review	Reported statistics; human-review controls hidden in UI.
PREPARED	100-paper pancreatic cancer corpus	Frozen curation is not a prospective discovery benchmark.
MEASURED	Standalone learned diagnostic	Synthetic null audit and one observational expression cohort.
TO PROVE	Useful discovery per dollar	No independent discovery, savings or customer result yet.

Source snapshot: main 1e83cfb, 6 September 2026. Recheck this ledger when new work lands.

Statistical evidence has a declared scope.

SYNTHETIC NULL • 10,000 RUNS / METHOD

Method	Ever crossed	Final reject
Identity	1.66%	0.13%
PCA	1.71%	0.14%
Learned encoder	1.12%	0.20%
Single feature	1.41%	0.36%
Fixed projection	1.65%	0.19%
Permutation p*	5.13%	5.13%
Calibrated permutation e*	0.24%	0.24%
Invalid memorization control	100.00%	100.00%

REAL-COHORT DIAGNOSTIC

19,000 → 512 → 128

Masked expression encoder
trained on separate donors

30 held-out pairs • observational cohort
Learned final e-value 3,652.81
PCA 85.76 • single gene 5,964.41

* Fixed-horizon comparators. No universal winner; lower null crossing does not establish greater power. Real cohort is COVID, not PDAC.

The pilot has explicit deployment gates.

COST	Meter model calls, compute, storage and scientist review. Cost per reviewed candidate = total pilot cost / reviewed candidates.
SECURITY	Authenticate research access; isolate experiment workers; scope credentials. Current localhost API and execution migration are not production-ready.
INTEGRATION	Keep corpus, source links, code and artifacts portable. Separate persistent workers and data from application releases.
ADOPTION & USE	Begin with nonclinical research decisions and public/licensed data. Private datasets and clinical use require separate data, legal and regulatory review.

Proposed gates. Price, gross margin and production compliance have not been measured or established.

The bar is a useful, inspectable research system.

Google DeepMind Co-scientist

Multi-agent hypothesis generation,
tools and scientist-led validation.

Strong evidence that multi-agent
research is already a serious field.

Elicit Research Agent

Research workflows, bioinformatics
analysis and source-linked outputs.

Paper summarization alone
is not a credible distinction.

Our focus

Grounded claim relationships →
recursive executable investigations.

Show exact provenance, decisions,
results and failure feedback.

Official product descriptions, accessed 6 September 2026. Differentiation is a design focus; comparative superiority is unproven.

The next result should be hard to fool.

01 / FIX THE TASK

Freeze the starting point

Version the corpus and data.
Choose tasks before runs.
Keep outcomes out of inputs.

02 / MATCH THE BUDGET

Compare real alternatives

Single-agent baseline.
Flat parallel research.
Our recursive controller.

03 / AUDIT THE OUTPUT

Count useful, reproduced work

Expert usefulness ratings.
Independent rerun success.
Time and total cost.

Proposed protocol; not run. Include failures and rejected branches. Model memory remains a concern in retrospective tasks.